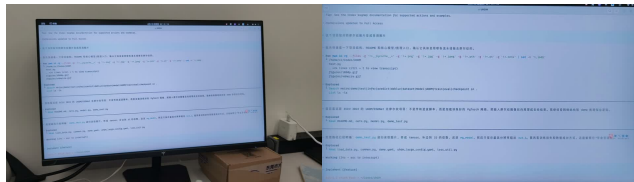


Screen Capture Rectification and Temporal Stabilization for Real-world Captured-screen Videos

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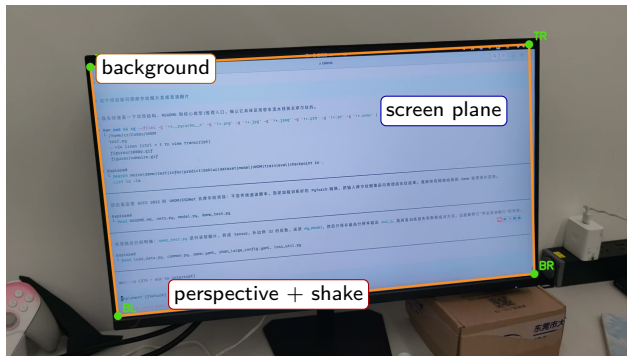
CIE6032 Final Project Proposal

June 24, 2026



Before restoration, OCR, or archiving, real filmed screens first need geometric normalization.

Motivation: real captured-screen videos are not pre-aligned



Gap

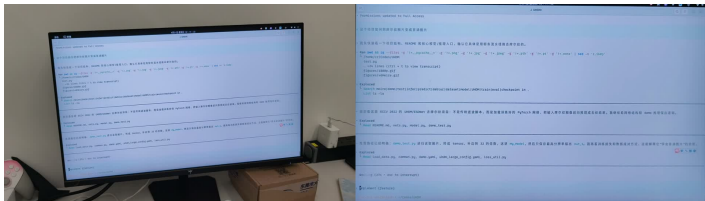
Many screen restoration settings start after the screen is already cropped or aligned.

- Real phone videos include background, perspective distortion, hand shake, weak borders, glare, moire, and dynamic content.
- We target the missing geometric preprocessing step before restoration.

full scene input

stable screen coordinates

Task: normalize a full camera video into screen coordinates



raw phone frame

rectified screen-coordinate output

Input

Hand-held video of a computer screen in a full scene.

Output

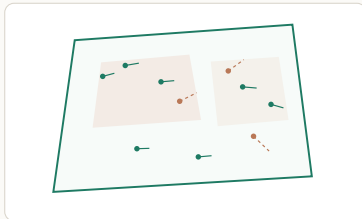
Front-facing screen-content video with aspect ratio preserved.

- Suppress background.
- Correct perspective.
- Reduce jitter.
- Preserve real content motion.

Method: estimate the screen plane, not moving inner content

Reference-plane Tracking

Use feature tracks that agree with a single screen-plane homography; reject dynamic-content outliers.



LEGEND

● RANSAC inliers

features consistent with screen plane

● Rejected outliers

dynamic content or tracking errors

□ Estimated plane

screen quadrilateral per frame

screen quadrilateral initialization

edge filtering +
LSD/Hough border lines

border-based ho-
mography candidate

LK inner points as
consistency check

RANSAC rejects con-
flicting content motion

rectify, smooth, or
re-detect/freeze

Current work: preliminary stability check

Current Work: Preliminary Stability Check

Pilot-clip p95 residual inter-frame motion after normalization. Lower values indicate more stable output.

Metric	Frame-wise Corner Detection	Optical Flow Tracking	PROPOSED Border-guided Tracking
Residual translation Full video, p95, px	2.49 px	2.44 px	0.146 px
Residual rotation Full video, p95, degree	0.0595 deg	0.0467 deg	0.0047 deg
Residual translation Last 2 s, p95, px	1.93 px	1.93 px	0.118 px
Residual rotation Last 2 s, p95, degree	0.0425 deg	0.0263 deg	0.0044 deg

Source: deliverables/proposal_20260622/evidence/proposal_ablation_summary.csv; one pilot captured-screen clip.
These are current-work stability diagnostics, not final dataset-wide reconstruction results.

Implemented so far

Detection, border-guided tracking, homography warping, and residual-motion analysis.

- Frame-wise detection is sensitive to corner jitter.
- Content optical flow can follow scrolling or in-screen video.
- Border-guided tracking is currently most stable on the pilot clip.

Pilot evidence only; full self-collected dataset evaluation is still planned.

Experiments: self-collected scenarios and baselines

Self-collected Evaluation Set

Final-stage plan: 5 scenario classes, 10 clips per class, approximately 5 seconds per clip.

50

captured-screen clips
balanced across realistic screen-content conditions

5s

short videos
enough for residual motion statistics and visual composition

5

classes
designed around failure modes

CLASS 1



Static page

documents, browser UI,
mostly fixed content
10 clips x 5s

CLASS 2



Scrolling page

page motion, cursor,
local UI changes
10 clips x 5s

CLASS 3



In-screen video

large dynamic regions
inside the display
10 clips x 5s

CLASS 4



PPT / weak border

low texture, pale UI,
unclear screen edges
10 clips x 5s

CLASS 5



4K / moiré / glare

hard cases for link to
restoration pipeline
10 clips x 5s

Dataset plan

5 scenario classes, 10 clips per class,
about 5 seconds per clip.

- Static pages.
- Scrolling pages.
- In-screen video playback.
- PPT or weak-border pages.
- Glare, moiré, or partial screen loss.

Baselines

Frame-wise detection, content optical flow, and proposed border-guided tracking.

Evaluation, expected result, and timeline

Aspect	Metrics
Geometry	corner error, quad IoU, aspect ratio error
Stability	residual translation, rotation, scale variation
Signal	gradient magnitude, edge preservation index
Grid/moire	2D FFT dominant direction check

Expected result

A classical geometric preprocessing pipeline that converts realistic captured-screen videos into stable, front-facing screen videos.

Date	Task
Jun. 22–24	finalize proposal and presentation
Jun. 25–26	collect 50 self-captured clips
Jun. 27–30	organize data and annotate key frames
Jul. 1–7	run ablations and metric evaluation
Jul. 8–10	prepare visual comparison and analysis
Jul. 11–15	finalize report, code, data, slides

geometry

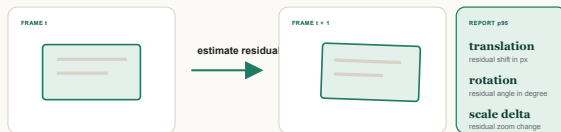
stability

signal

Backup: evaluation metric visual

Residual Motion Metrics

We evaluate stability by estimating the residual affine motion between adjacent normalized frames.



Interpretation

Lower p95 residual motion means the normalized output is more stable. These are stability metrics, not ground-truth image reconstruction scores.

- Annotated corners measure geometric correctness.
- Adjacent-frame affine residuals measure stability.
- Gradient, edge, and FFT checks measure signal preservation after rectification.

Backup: expected failure cases for QA

Failure Modes to Analyze

Failure examples are evidence: they define the operating boundary of the preprocessing chain.

CASE 1

Weak border / low texture

PPT and white pages may not provide reliable corners or stable feature points.



CASE 2

Scrolling content drift

Content motion can dominate optical flow and pull the estimated homography away.



CASE 3

In-screen video motion

Large dynamic areas reduce the fraction of features tied to the physical screen plane.



CASE 4

Moiré / glare hard cases

Moiré, reflections, and saturation affect both screen detection and feature tracking.



Why not directly train demoireing?

This project targets the earlier geometric stage needed before restoration.

How handle dynamic content?

Border motion drives the homography; inconsistent inner LK motion is rejected as content motion.